

(15 pts): Weiss, Exercise 9.15

- a. Find a minimum spanning tree for the graph in Figure 9.84 using both Prim's and Kruskal's algorithms.
- b. Is this minimum spanning tree unique? Why?

A. See 915.jpg

Prim's Order Settled: A - (A,B) B - (B,E) E - (E,I) H - (E,H) I - (I,F) F - (F,G) G - (G,C) C - (G,C) D - (A,D) J - (G,J)

B. Not a unique MST since there is a choice of edges to be made for both Prim's Algorithm and Kruskal's Algorithm. This will lead to different MSTs.

(15 pts): Weiss, Exercise 9.53 - Hint: Start with one of the graph algorithms we discussed in class.

The object of the Kevin Bacon Game is to link a movie actor to Kevin Bacon via shared movie roles. The minimum number of links is an actor's Bacon number. For instance, Tom Hanks has a Bacon number of 1; he was in Apollo 13 with Kevin Bacon. Sally Field has a Bacon number of 2, because she was in Forrest Gump with Tom Hanks, who was in Apollo 13 with Kevin Bacon. Almost all well-known actors have a Bacon number of 1 or 2. Assume that you have a comprehensive list of actors, with roles, and do the following:

- a. Explain how to find an actor's Bacon number.
 - b. Explain how to find the actor with the highest Bacon number.
 - c. Explain how to find the minimum number of links between two arbitrary actors.
- a. First we would need to match actors based on movies with shared roles. This could be done with Kruskal's algorithm using Movies to represent edges, and Actors as vertices. This would give us an MST of all actors and their degrees of separation from each other. To find an actor's Bacon number, we would then need to find the path from a Kevin Bacon to any other vertex and count the number of edges traversed.
- b. Here we could create a spanning tree from a graph of Actors (Vertices) and Shared Movies (edges). We could then perform an iterative Depth-First Search and get the distance between the root and the first visited neighbor.
- c. This is basically the same process as a, except using two different actors instead of KB and 1 other.